

Symbolic-Aware Music Understanding and Generation with Large Audio Language Models

Dabin Kim, Minhee Lee, Ji Jiaxian, Aner Zheng

Graduate School of Cultural Technology, KAIST, Korea

Motivation

Why Symbolic Representation Matters for Music Understanding and Generation?

- Audio tokens capture sound, but not always musical structure.
- Music tasks often depend on discrete structure: **beats**, **chords**, **notes**, **sections**.
- We need to test symbolic gains, not just extra data or extra tokens.

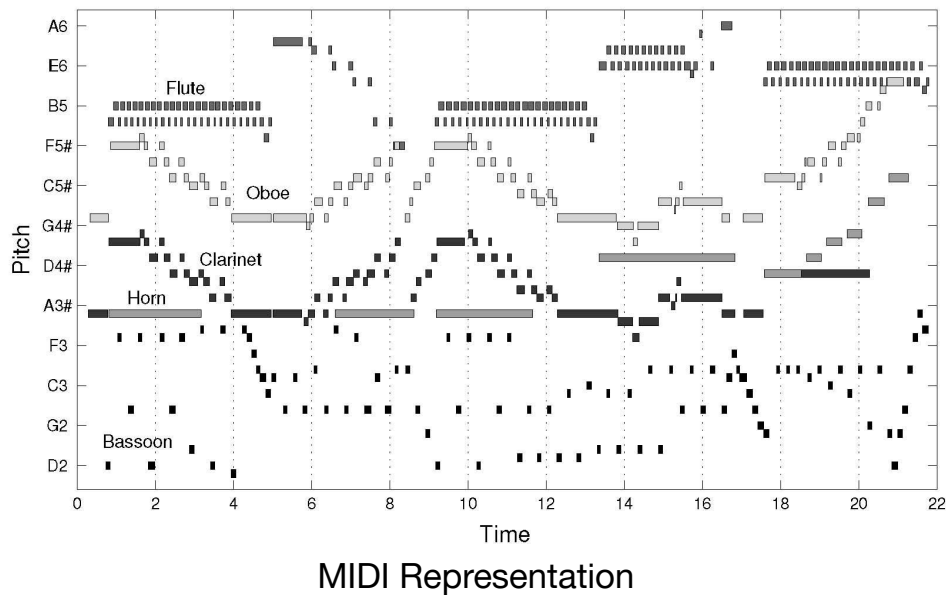


Figure 1.1 from [Müller, FMP, Springer 2015]

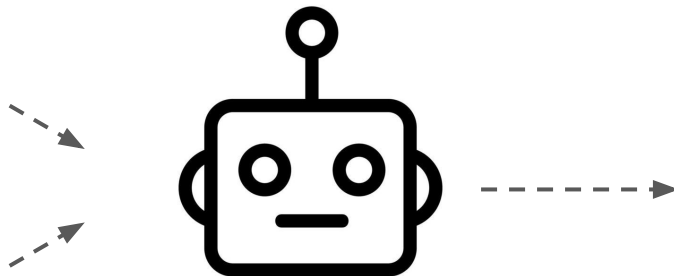
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“Please tell me the melodic progression
of this bass instrumental audio.”



“...?”

Why Add Symbolic Understanding?

Chord progression, stem-level instrumentation, phrase-level reasoning,

Symbolic QA, MIR probing, audio-symbolic retrieval

Chord / melody / rhythm / instrumentation control

4

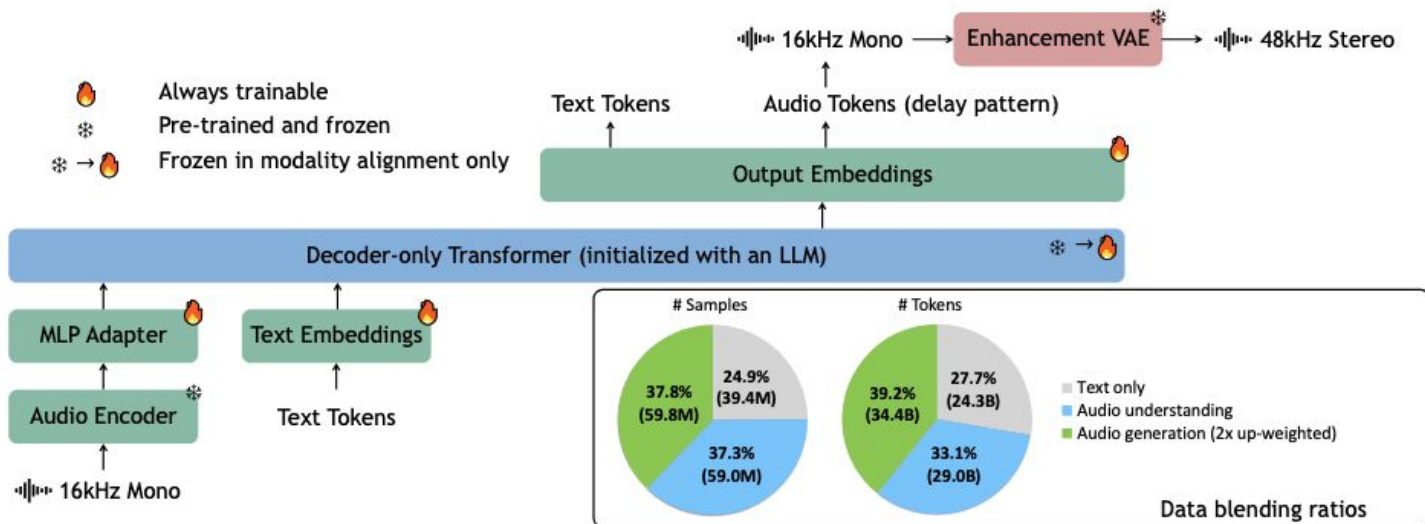
Related Works: UALM

Architecture: Continuous Input, Discrete Output

- Base: Qwen2.5 decoder-only LLM.

Audio input: acoustic encoder > continuous embeddings > MLP adapter.

- Audio output: X-codec tokens at 50 Hz, 8 RVQ tokens/frame.
- Enhancement VAE upgrades 16 kHz mono to 48 kHz stereo.



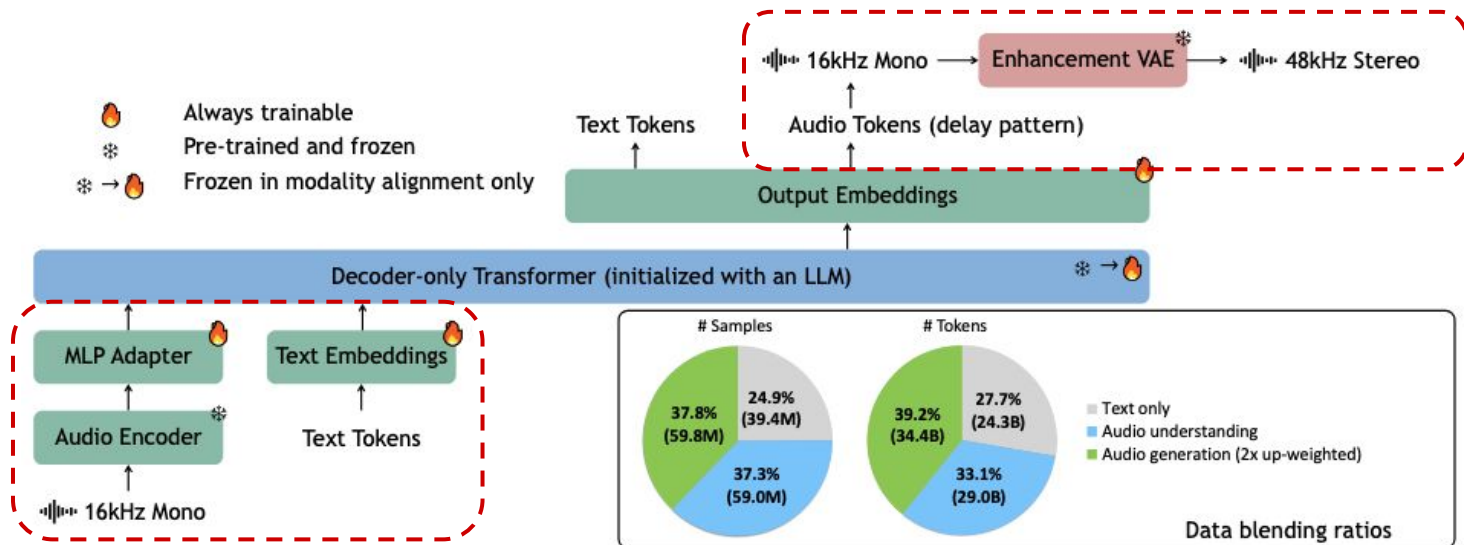
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Training Recipe for LM-Based Audio Generation

- UALM-Gen uses about 30M text-audio pairs, around 80k hours.
- Generation data is up-weighted because audio-token convergence is slow.
- Delay pattern reduces multi-codebook sequence burden.
- CFG and self-adaptation + DPO improve prompt adherence and quality.

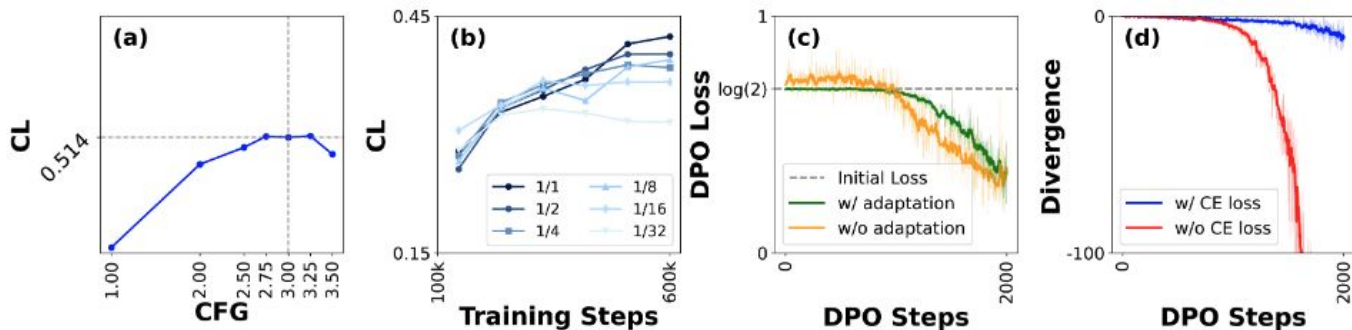


Figure 5: Statistics of UALM-Gen model. (a) The CLAP scores (CL) with various CFG λ ; (b) The CLAP scores (CL) with various training data volume down-weighting; (c) the DPO loss w/o adaptation on synthetic data before DPO training; (d) the divergence $\pi_{\theta}(y_w|x) - \pi_{\text{ref}}(y_w|x)$ from the reference model w/o CE loss added in DPO training.

Related Works: UALM

UALM-Reason: Understanding-Generation Loop

- Rich caption is a generation blueprint: keywords, layout, description.
- Enrichment: short prompt -> detailed generation plan.
- Dialogue: ask for missing details before generation.
- Self-reflection: generate -> listen again -> critique -> refine.

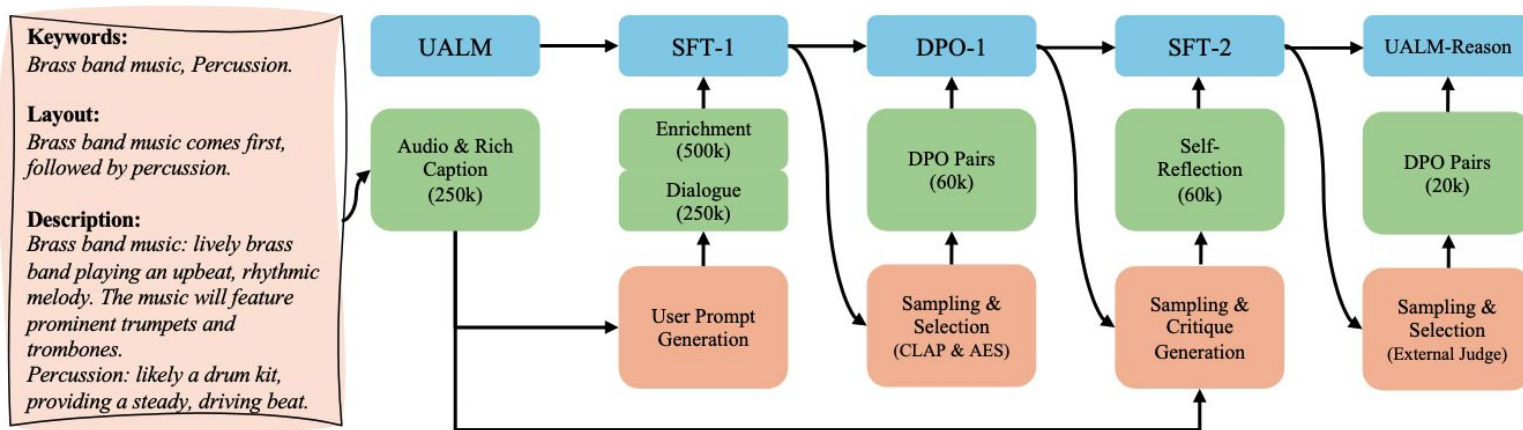


Figure 3: Rich caption example (left) and post-training workflow (right).

Related Works: UALM

Results and Caveats

- Understanding: MMAU mean 74.1, competitive with Qwen2.5-Omni and Audio Flamingo 3.
- Generation: LM-based UALM-Gen competes with strong diffusion baselines.
- Text reasoning is mostly preserved after multimodal pretraining.
- Reasoning gains are mainly evaluated with subjective and qualitative evidence.

Model	Base Model	MMAU-v05.15.25				MMAR
		Sound↑	Music↑	Speech↑	Mean↑	Mean↑
GAMA-IT (Ghosh et al., 2024)	LLaMA-2 (7B)	32.7	22.4	11.6	22.2	17.4
SALMONN (Tang et al., 2024)	Vicuna (13B)	42.1	37.8	28.8	36.2	33.2
Qwen2-Audio-Instruct (Chu et al., 2024)	Qwen2-Audio (7B)	61.2	55.7	55.4	57.4	30.0
DeSTA2.5-Audio (Lu et al., 2025)	Llama3.1-8B-Instruct	66.8	57.1	<u>71.9</u>	65.2	—
Audio Reasoner (Xie et al., 2025)	Qwen2-Audio Instruct (7B)	67.3	61.5	62.5	63.8	36.8
Step-Audio-2 (Wu et al., 2025a)	Unknown (<130B)	80.6	68.2	72.8	<u>73.9</u>	—
Qwen2.5-Omni (Xu et al., 2025)	Qwen2.5 (7B)	76.8	67.3	68.9	71.0	<u>56.7</u>
Audio Flamingo 3 (Goel et al., 2025)	Qwen2.5 (7B)	76.7	<u>73.3</u>	64.9	72.3	58.5
UALM (Ours)	Qwen2.5 (7B)	<u>77.9</u>	77.6	66.7	74.1	55.2

Comparison with Understanding Models

Related Works: UALM

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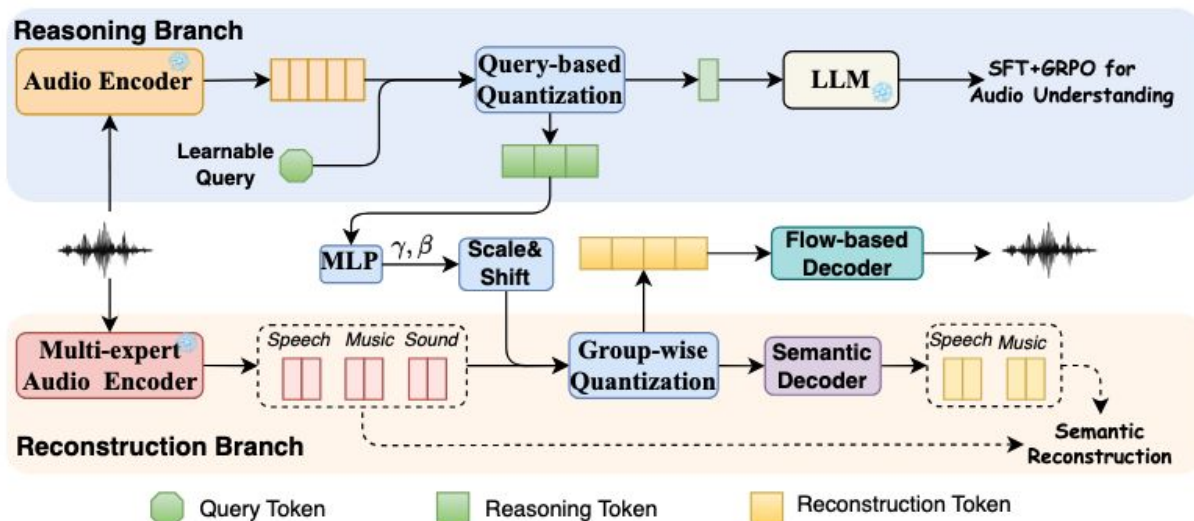
Model	SongDescriber							AudioCaps						
	FD↓	KL↓	IS↑	CL↑	AES↑	OVL↑	REL↑	FD↓	KL↓	IS↑	CL↑	AES↑	OVL↑	REL↑
Ground Truth	0	0	1.88	0.48	7.20	4.10	4.03	0	0	13.49	0.62	4.50	3.91	3.96
MusicGen-stereo-L (Copet et al., 2024)	228.94	0.84	1.69	0.36	6.64	3.91	<u>3.97</u>	—	—	—	—	—	—	—
AudioGen-M (Kreuk et al., 2022)	—	—	—	—	—	—	—	149.70	1.35	12.26	0.57	4.21	3.83	3.89
MAGNeT-M (Ziv et al., 2024)	191.49	0.70	1.47	0.41	6.65	3.84	3.86	149.68	1.86	7.73	0.46	4.08	3.68	3.72
AudioLDM2-L (Liu et al., 2024b)	331.73	0.68	1.96	0.45	6.31	3.83	3.80	121.63	1.72	8.59	0.52	4.31	<u>3.85</u>	3.75
TangoFlux (Hung et al., 2024)	235.61	0.71	1.70	0.41	6.46	3.80	3.89	103.04	1.02	<u>15.13</u>	0.65	4.42	3.72	<u>3.93</u>
Stable Audio Open (Evans et al., 2024)	138.58	1.01	2.25	0.42	6.37	3.92	<u>3.97</u>	100.93	2.22	11.80	0.35	4.47	3.81	3.80
ETTA (Lee et al., 2024)	95.66	0.80	<u>2.15</u>	0.44	6.71	3.92	3.93	80.13	1.22	14.36	0.54	4.51	3.73	3.94
UALM-Gen (Ours)	74.43	<u>0.63</u>	1.87	0.54	7.36	4.07	3.96	<u>75.14</u>	<u>1.19</u>	14.52	0.65	5.08	3.79	3.92
UALM (Ours)	<u>83.69</u>	0.59	2.00	0.54	<u>7.28</u>	<u>3.97</u>	3.99	65.87	1.35	15.62	0.62	<u>4.92</u>	3.89	3.86

Comparison with Generative Models

Related Works: UniAudio 2.0

ReasoningCodec: Factorized Audio Tokens

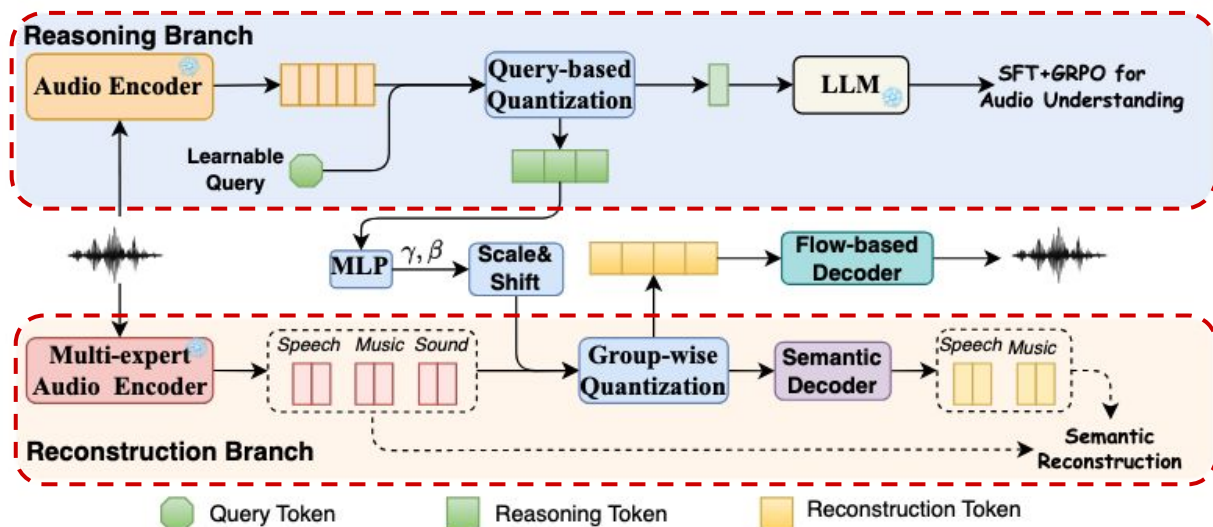
- Audio is split into reasoning tokens and reconstruction tokens.
 - **Reasoning tokens:** low-rate, language-aligned, high-level abstraction.
 - **Reconstruction tokens:** acoustic detail for reconstruction and generation.
- **Core Idea:** understanding and generation need different information.



Related Works: UniAudio 2.0

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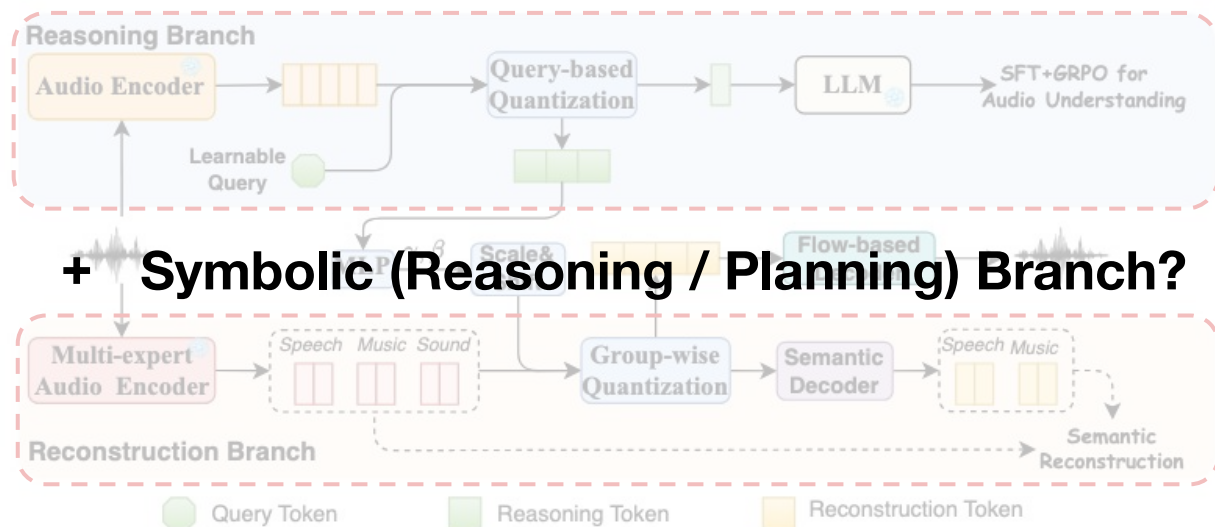
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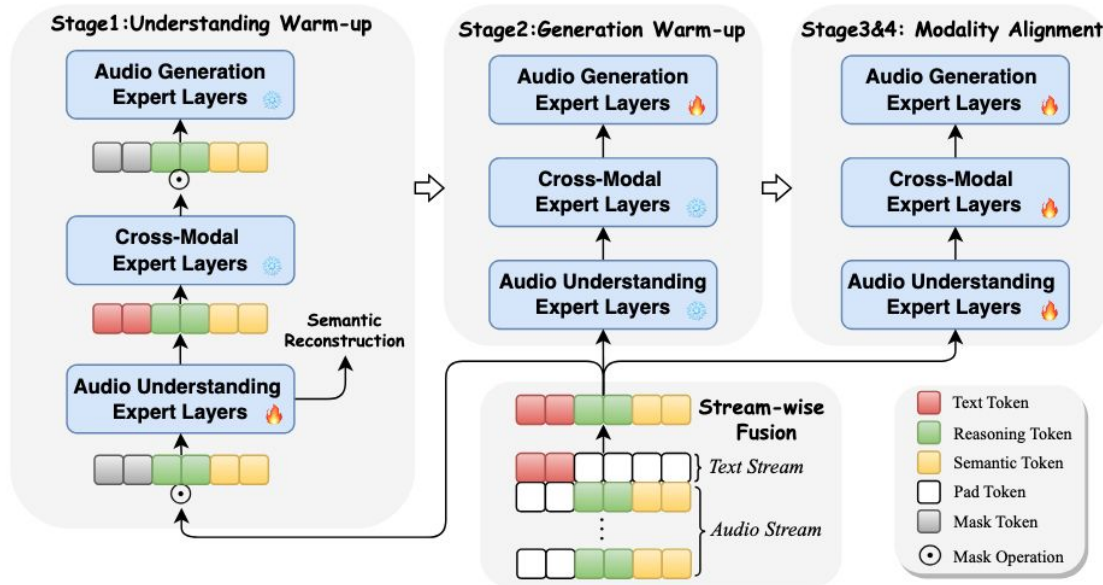
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How Reasoning Tokens Are Trained

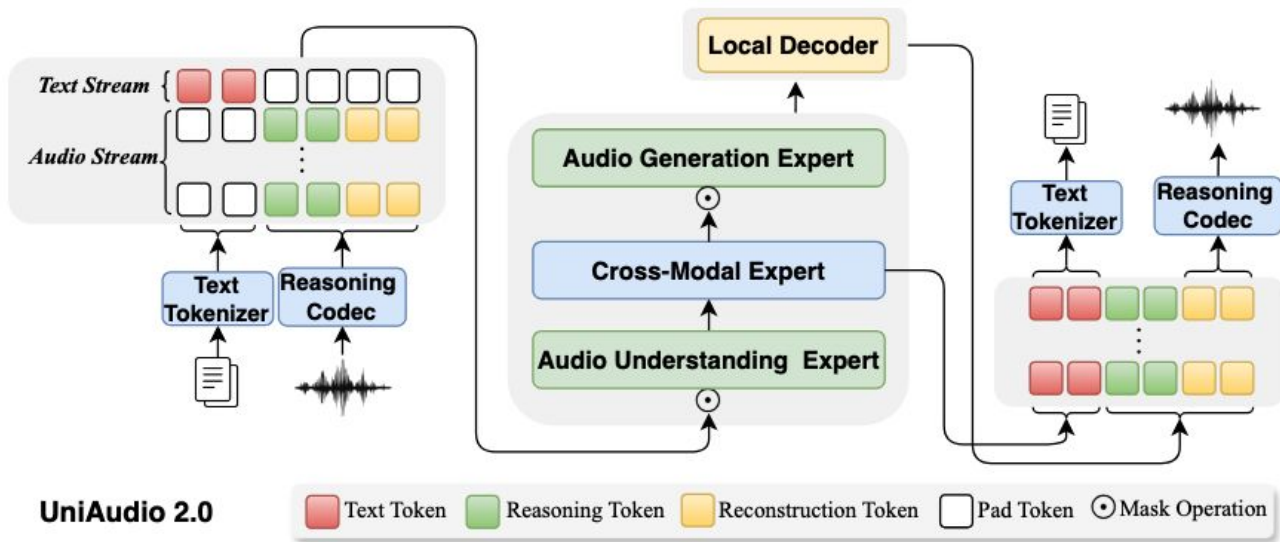
- SFT uses ASR, captioning, classification, and audio reasoning tasks.
- Qwen3-Omni and Gemini 2.5 Pro help construct reasoning supervision.
- GRPO uses rule rewards for verifiable tasks and LLaMA-3.1-Instruct-8B judge for fine-grained analysis.
- Reconstruction branch is trained after freezing the reasoning branch.



Related Works: UniAudio 2.0

Unified AR Model: 9-Stream Representation

- One sequence packs text, reasoning tokens, and reconstruction tokens.
- Audio uses 8 codebook streams; text uses 1 stream.
- Stream-wise embeddings are summed into one hidden vector.
- Audio-specific experts update audio positions only.



Related Works: UniAudio 2.0

Results, Strengths, and Gaps

Table 3. Downstream understanding evaluation on LLM-based ASR (ASR), emotion classification (ER), audio classification (AC), and music classification (MC).

Model	ASR	ER	AC	MC
Whisper	8.5	59.2	54.3	74
DAC	93.2	5.2	19	28
XCodec	37.6	29.4	44.7	46
Higg-Codec	31.2	30	49.4	38
ALMTokenizer	26.9	32.4	50.1	45
Reason-only	10.1	50.2	33	80
Reconstruction-only	16.3	42.1	48.7	65
Reason+Reconstruction	9.0	56.4	63.3	70

Related Works: UniAudio 2.0

Results, Strengths, and Gaps

Table 5. The performance comparison on Seen tasks between previous SOTA models and UniAudio 2.0. For each task, we choose the commonly used benchmark and metrics. Note that we list the most representative and related works with us, the more comprehensive comparison can be found in Appendix B.6. **Bold** for the best result.

Task & Datasets	Model	Performance
TTS ZH / EN / LS-clean	MiMo-Audio-7B-Instruct	1.93 / 5.37 / 4.74
	Qwen2.5-Omni 7B	1.21 / 3.10 / 4.28
	UniAudio 2.0 (Ours)	2.30 / 3.63 / 3.46
InstructTTS WER/ Style-Acc / UTMOSv2	MiMo-Audio-7B-Instruct	7.8 / 40.5 / 3.17
	CapSpeech-AR	9.1 / 52.2 / 3.18
	UniAudio 2.0 (Ours)	7.3 / 42.3 / 3.38
ASR LS-clean / LS-other / Seed-EN / Seed-ZH	MiMo-Audio-7B-Instruct	3.5 / 35.4 / 29.8 / 7.0
	Qwen2.5-Omni-7B	3.9 / 5.5 / 1.3 / 2.9
	UniAudio 2.0 (Ours)	2.7 / 6.3 / 2.6 / 2.1
Audio Caption & Generation CIDER / KL / FD	Qwen2.5-Omni	0.39 / - / -
	Stable Audio Open	- / 2.14 / 78.2
	UniAudio 2.0	0.69 / 3.26 / 50.7
Music Caption & Generation GPT-score / KL / FAD	Qwen2.5-Omni-7B	5.33 / - / -
	MusicGen	- / 1.31 / 5.0
	UniAudio 2.0	5.14 / 1.8 / 3.44
Song Generation WER / CE/CU/PQ	SongGen	40.58 / 6.77 / 6.86 / 7.19
	UniAudio 2.0	36.5 / 6.87 / 7.41 / 7.62
Lyric Recognition WER	Qwen2.5-Omni-7B	56.99
	UniAudio 2.0	28.57

Related Works: UniAudio 2.0

Results, Strengths, and Gaps

- ReasoningCodec improves tokenizer-level understanding and reconstruction.
- SFT + GRPO improves ASR, classification, relevance, and fluency.
- Music generation is competitive, but Non-MIR metric-dependent.
- Music captioning is not clearly strongest against Qwen2.5-Omni / Audio Flamingo 3.

Research Gap

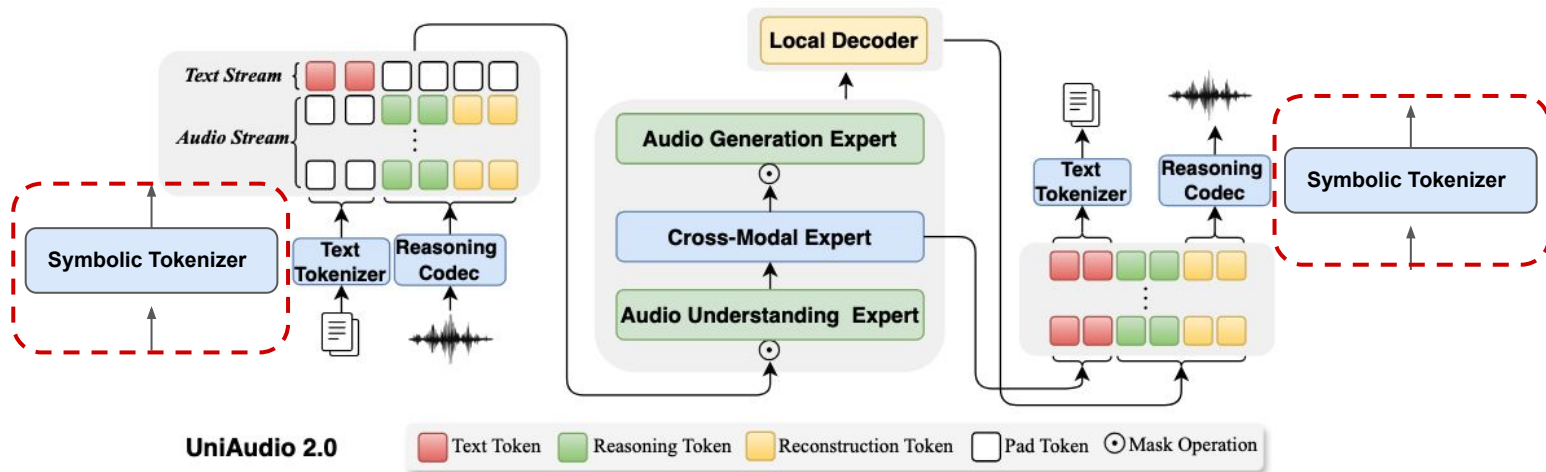
What Remains Unresolved for This Project

- UALM addresses the process axis: planning, generation, self-evaluation.
- UniAudio 2.0 addresses the representation axis: reasoning vs reconstruction tokens.
- The missing axis is explicit music-symbolic grounding: chord, meter, phrase, MIDI/score-level control.
 - Shuffled-symbolic token controls are needed before claiming causal use of structure.

Problem Definition

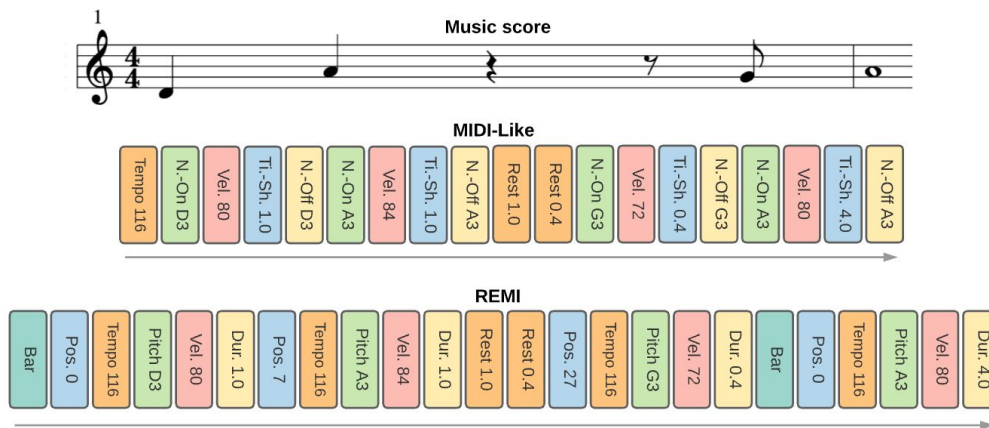
Key Goals

- Add a learned symbolic stream to a UniAudio-style LALM.
- Show gains in music controllability and structural understanding.
- Separate symbolic gains from extra data, tokens, and parameters.



Proposed Approach

- Add a Learned Symbolic Planning Stream
- **Base sequence:** text + Ra + Ca
- **Proposed sequence:** text + Ra + Ca + S-plan
 1. **S-surface** : normalized symbolic events.
 2. **S-plan** : low-rate learned symbolic planning token.
- **Desired:** low-rate, grounded, complementary, editable



Example of Basic Symbolic Tokenization (MIDITok)

Dataset (Expected)

- MIDI-to-Audio synthesized paired data for the scalability precise temporal alignment.
- 1. Slakh2100 [1]**
 - a. Synthesized with professional-grade sample-based virtual instruments.
 - b. Contains 2100 automatically mixed tracks and accompanying, aligned MIDI files, synthesized from 187 instrument patches categorized into 34 classes.
 - 2. AAM: Artificial Audio Multitracks Dataset [2]**
 - a. 3,000 artificial music audio tracks with rich annotations.
 - b. The annotation files include: Onsets, Pitches, Instruments, Keys, Tempos, Segments, Melody instrument, Beats, and Chords.

[1] Slakh2100: Manilow et al., WASPAA 2019, arXiv:1909.08494.

[2] AAM: Ostermann et al., EURASIP JASMP 2023, DOI:10.1186/s13636-023-00278-7.

Evaluation Methods (Expected)

1. Token-Level Causal Evaluation

- a. Token causality
- b. Token health

2. Benchmark-First Evaluation

- a. CLaMP3 [1]-style retrieval (audio-symbolic alignment)
- b. MTG/MagnaTagATune-style AUC/AP/F1 (music tagging)
- c. MIR Eval chord / beat / onset / transcription metrics (musical structure)

3. QA Control Protocol

- a. RUListening [2]
- b. MuChoMusic [3]

[1] Source: Wu et al., “CLaMP 3,” ACL 2025, arXiv:2502.10362.

[2] Zang et al., 2025, arXiv:2504.00369.

[3] Weck et al., ISMIR 2024, arXiv:2408.01337.

Summary

Final Thesis

1. Symbolic understanding should improve music understanding.
2. It should expand musical downstream tasks beyond captioning/generation.
3. It should enable symbolic-based controllable generation.

The central claim is not “add MIDI tokens,” but “prove when symbolic planning information changes what a music LALM can understand, evaluate, and control.”